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PROJECT REPORT

Fit Hub CRM - Fitness Centre Management System

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ABSTRACT

The fitness industry has witnessed exponential growth over the past decade, with an increasing number of individuals subscribing to gym memberships, personal training sessions, and wellness programmes. However, most fitness centres continue to rely on manual processes or fragmented tools to manage their day-to-day operations, resulting in inefficiencies, data inconsistencies, and poor member experiences.

FitHub CRM is a web-based Fitness Centre Management System designed to address these operational challenges through a centralised, role-based Customer Relationship Management (CRM) platform. The system integrates four primary modules: Member Management, Trainer Management, Attendance Tracking, and Payment & Subscription Management, enabling fitness centre administrators to manage all aspects of their business from a single unified interface.

The application is developed using React.js for the front-end interface, Node.js with Express.js for the back-end server, and MySQL for the relational database. The system supports role-based access control, allowing administrators, trainers, and members to interact with the platform based on their respective privileges.

Key features include automated membership renewal alerts, real-time attendance marking via QR code or manual entry, trainer scheduling, payment history tracking, and a dashboard with operational analytics. The system is designed to be responsive and accessible across desktop and mobile devices.

The implementation follows Agile methodology, with development broken into structured sprints covering requirement gathering, design, module development, integration, and testing phases. The project aims to improve administrative efficiency, enhance member engagement, and provide fitness centre owners with data-driven insights to support business growth.

Keywords: CRM, Fitness Management System, Member Management, Trainer Scheduling, Attendance Tracking, Subscription Management, React.js, Node.js, MySQL.

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CHAPTER 1: INTRODUCTION

1.1 Overview of the Fitness Industry

The global fitness industry has grown into a multi-billion-dollar sector, driven by increasing health consciousness, urbanisation, and changing lifestyle patterns. India's fitness market has witnessed remarkable expansion, with thousands of fitness centres, gyms, and wellness studios emerging in tier-1 and tier-2 cities. With the growing adoption of structured workout regimens, diet plans, and personalised training programmes, fitness centres have evolved from simple exercise facilities into comprehensive wellness ecosystems.

Despite this growth, a significant portion of fitness centres, particularly small and medium-sized businesses, continue to rely on manual methods for managing memberships, tracking attendance, and processing payments. These approaches are not only time-consuming but also susceptible to errors, making it difficult to scale operations and maintain consistent service quality. There is a pressing need for a technology-driven management solution that can streamline operations and improve the overall member experience.

Customer Relationship Management (CRM) systems have long been used in industries such as retail, healthcare, and banking to centralise customer data, automate workflows, and improve service delivery. Applying CRM principles to the fitness industry presents a significant opportunity to modernise how fitness centres manage their relationships with members and trainers.

1.2 Problem Statement

Fitness centres face several operational challenges that hinder their growth and efficiency. The following are the key problems identified through research and stakeholder discussions:

- Member data is stored in physical registers or unstructured spreadsheets, making retrieval slow and error-prone.
- Attendance tracking is manual and inconsistent, with no reliable mechanism to monitor member visit frequency or trainer availability.
- Payment and subscription management lacks automation, resulting in delayed renewals, missed follow-ups, and revenue leakage.
- Trainer schedules are managed informally, leading to conflicts, double bookings, and poor utilisation of trainer time.
- There is no centralised platform for administrators to view business metrics such as active members, pending payments, or trainer performance.
- Members have no self-service portal to view their schedule, track their progress, or update their profile.

These challenges collectively reduce operational efficiency, impact member satisfaction, and limit the ability of fitness centre owners to make data-driven business decisions.

1.3 Objectives

The primary objectives of the FitHub CRM system are as follows:

1. To develop a centralised web-based platform for managing all fitness centre operations including member registration, trainer assignment, attendance, and payments.
2. To implement role-based access control, enabling distinct dashboards and functionalities for administrators, trainers, and members.
3. To automate membership renewal notifications and payment reminders, reducing manual follow-up efforts.
4. To enable real-time attendance tracking for members and trainers using a reliable and easy-to-use interface.
5. To provide an administrative dashboard with key performance indicators (KPIs) such as active members, monthly revenue, and trainer utilisation.
6. To build a responsive and user-friendly interface accessible across desktop and mobile devices.
7. To ensure data security through user authentication, encrypted passwords, and session management.

1.4 Scope of the Project

The FitHub CRM project encompasses the design, development, and testing of a web-based fitness centre management system. The scope of the project includes the following areas:

Inclusions:

- Member registration, profile management, and membership plan assignment.
- Trainer profile creation, schedule management, and batch assignment.
- Daily attendance marking with historical records for both members and trainers.
- Payment recording, subscription tracking, and automated renewal alerts.
- Administrative dashboard with graphical reports and data summaries.
- Member self-service portal for viewing schedules and personal records.

Exclusions:

- Integration with third-party payment gateways (future scope).
- Mobile application development (Android/iOS).
- Nutritional tracking and diet planning modules.
- Integration with wearable fitness devices.

CHAPTER 2: IDEATION PHASE

2.1 Problem Statement Table

The following table presents a structured breakdown of the problems identified, their root causes, and the impact on stakeholders in the fitness centre ecosystem.

Problem	Root Cause	Impact	Affected Stakeholder
Manual membership records	No digital registration system	Data loss, slow retrieval	Admin, Members
Irregular attendance tracking	Paper-based registers	Inaccurate reports	Admin, Trainers
Missed payment renewals	No automated reminders	Revenue loss	Admin, Members
Trainer scheduling conflicts	Verbal/manual scheduling	Poor trainer utilisation	Trainers, Members
Lack of business insights	No dashboard or analytics	Poor decision-making	Admin / Owner
No member self-service	No member-facing portal	Low member engagement	Members

2.2 Empathy Map

An Empathy Map was developed to understand the perspectives of the key stakeholders. The map captures what users Say, Think, Do, and Feel in the current state of fitness centre management.

Stakeholder: Gym Administrator

Says	Thinks	Does	Feels
"I have to check three different sheets just to find one member's status."	Why isn't there a single place for everything?	Maintains manual registers, calls members for renewals	Frustrated, overwhelmed

Stakeholder: Fitness Trainer

Says	Thinks	Does	Feels
"I don't know which members are in my batch today."	There should be a proper schedule I can refer to.	Asks admin verbally every morning	Confused, unprepared

Stakeholder: Gym Member

Says	Thinks	Does	Feels
"I had no idea my membership expired last week."	I wish I could check my status online.	Visits the gym without knowing expiry status	Embarrassed, inconvenienced

2.3 Brainstorming

The team conducted a structured brainstorming session to identify potential solutions to the problems identified. The following ideas were explored:

Ideas Generated:

- Mobile application with QR code-based attendance scanning.
- Integration with biometric devices for automated check-in.
- WhatsApp-based chatbot for membership renewals and reminders.
- Web-based CRM platform with role-based dashboards.
- Spreadsheet automation using Google Sheets with custom scripts.
- Cloud-hosted ERP system with fitness centre-specific modules.

Evaluation Criteria:

Idea	Feasibility	Cost	Scalability	User Friendliness	Selected
Mobile App with QR	Medium	High	High	High	No
Biometric Integration	Low	Very High	Medium	Medium	No
WhatsApp Chatbot	Medium	Medium	Low	High	No
Web-based CRM	High	Low	High	High	Yes
Google Sheets Scripts	High	Low	Low	Medium	No
Cloud ERP	Low	Very High	High	Medium	No

2.4 Selected Idea

After evaluating all proposed solutions against the defined criteria, the team selected the Web-Based CRM Platform as the most appropriate solution. This approach offers the best balance of feasibility, affordability, scalability, and user-friendliness.

The selected idea — FitHub CRM — is a role-based, web-based fitness centre management system that centralises all operational functions within a single, secure platform. The system will be built using modern web technologies including React.js, Node.js, and MySQL, enabling rapid development, ease of maintenance, and a responsive user experience across devices.

This solution directly addresses all identified pain points without requiring expensive hardware (such as biometric devices) or dependency on third-party platforms (such as WhatsApp). It can be deployed on a shared cloud server, keeping infrastructure costs minimal for small and medium-sized fitness centres.

CHAPTER 3: REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The Customer Journey Map traces the experience of a Gym Member from initial registration to active usage of the FitHub CRM system. It highlights key touchpoints, emotions, and opportunities for system intervention.

Stage	Member Action	Touchpoint	Pain Point	CRM Solution
Awareness	Enquires about gym membership	Front desk / phone	No online info available	Member portal with plan details
Registration	Fills registration form	Paper form / admin entry	Slow, error-prone process	Digital online registration
Onboarding	Assigned to trainer & batch	Verbal communication	No written schedule given	Auto-assigned batch with notification
Active Use	Attends gym daily	Physical attendance register	Inconsistent tracking	Digital attendance marking
Renewal	Membership nearing expiry	Admin calls / no reminder	Often missed, lapses occur	Automated email/SMS reminder
Payment	Pays membership fee	Cash payment at desk	No digital receipt/record	Online payment recording with receipt

3.2 Functional Requirements

The functional requirements define the specific behaviours and capabilities that the FitHub CRM system must exhibit. These are categorised by module.

Member Management Module:

8. The system shall allow administrators to register new members with details including name, contact information, date of joining, and selected membership plan.
9. The system shall allow administrators to view, edit, and deactivate member profiles.
10. The system shall display membership expiry status and classify members as Active, Expiring Soon, or Expired.

Trainer Management Module:

11. The system shall allow administrators to register trainers with their specialisation, contact details, and assigned batches.
12. The system shall enable batch creation with defined time slots and trainer assignments.
13. The system shall allow trainers to view their assigned batches and member lists.

Attendance Module:

14. The system shall allow daily attendance marking for members and trainers.
15. The system shall maintain a historical log of attendance records, filterable by date and member.
16. The system shall generate attendance summary reports for individual members.

Payment Module:

17. The system shall record payment transactions for membership subscriptions.
18. The system shall generate and display payment receipts for each transaction.
19. The system shall send automated alerts when a membership is within 7 days of expiry.

Dashboard Module:

20. The system shall provide an administrative dashboard displaying total active members, pending payments, trainer count, and daily attendance summary.
21. The system shall display graphical reports for monthly revenue and member growth trends.

3.3 Non-Functional Requirements

The non-functional requirements define the quality attributes that the FitHub CRM system must satisfy.

Category	Requirement	Specification
Performance	System response time	All pages must load within 3 seconds under normal load
Availability	System uptime	System should be available 99% of the time during business hours
Security	Authentication	All users must authenticate with encrypted credentials (bcrypt)
Security	Session Management	Sessions must expire after 30 minutes of inactivity
Scalability	Concurrent users	System must support at least 100 concurrent users without performance degradation

Category	Requirement	Specification
Usability	Interface design	System must be operable by non-technical staff with minimal training
Compatibility	Browser support	Compatible with Chrome, Firefox, and Safari (latest 2 versions)
Reliability	Data backup	Automated daily database backups must be maintained
Maintainability	Code quality	Source code must be modular, commented, and version-controlled

3.4 Data Flow Description

The data flow within the FitHub CRM system is structured across three primary layers: the Presentation Layer (front-end), the Application Layer (back-end API), and the Data Layer (database). The following describes the key data flows:

Member Registration Flow:

22. Administrator enters member details in the registration form (Presentation Layer).
23. Form data is validated on the client side and submitted via an HTTP POST request to the Node.js API (Application Layer).
24. The API validates the data, hashes the password if applicable, and inserts the record into the Members table (Data Layer).
25. A success response is returned and the administrator is redirected to the member list page.

Attendance Marking Flow:

26. Trainer or administrator accesses the attendance module and selects the date and batch.
27. Member list is retrieved from the database and displayed.
28. Administrator marks attendance (Present/Absent) and submits.
29. Attendance records are stored with timestamps in the Attendance table.

Payment Processing Flow:

30. Administrator records payment details including member ID, amount, plan, and date.
31. Payment record is saved to the Payments table and linked to the member's subscription.
32. Membership expiry date is automatically updated based on the selected plan.
33. System checks for upcoming expiries daily and queues reminder notifications.

3.5 Technology Stack

Layer	Technology	Purpose
Front-End	React.js	Single-page application with component-based UI
Front-End	Bootstrap 5 / CSS3	Responsive layout and styling
Front-End	Axios	HTTP client for API communication
Back-End	Node.js	Server-side JavaScript runtime environment
Back-End	Express.js	RESTful API framework
Back-End	JWT (JSON Web Token)	Stateless authentication and session management
Back-End	bcrypt.js	Password hashing and security
Database	MySQL	Relational database management system
Database	Sequelize ORM	Database abstraction and query management
Development Tools	VS Code	Primary integrated development environment (IDE)
Development Tools	Git / GitHub	Version control and collaboration
Deployment	Apache / Nginx	Web server for production deployment

CHAPTER 4: PROJECT DESIGN

4.1 Problem-Solution Fit

The Problem-Solution Fit validates that the proposed FitHub CRM system directly addresses each of the identified problems with a concrete, implementable feature.

Problem	Proposed Solution	Module
Manual membership records	Digital member registration with structured database	Member Management
Inconsistent attendance tracking	Daily digital attendance marking with historical logs	Attendance Module
Missed payment renewals	Automated expiry alerts and payment history dashboard	Payment Module
Trainer scheduling conflicts	Structured batch creation with trainer assignment	Trainer Management
No business insights	Administrative KPI dashboard with graphical reports	Dashboard
No member self-service	Member login portal with profile and schedule view	Member Portal

4.2 Proposed Solution

FitHub CRM is a comprehensive, web-based Customer Relationship Management system tailored specifically for fitness centres. The system is designed around four core operational pillars:

Membership Management:

The system provides a structured digital workflow for member onboarding. Upon registration, each member is assigned a unique ID, linked to a membership plan (monthly, quarterly, or annual), and assigned to a trainer and batch. The system maintains a complete profile for each member including personal details, contact information, emergency contacts, health declarations, and membership history.

Trainer and Batch Management:

Trainers are onboarded into the system with their specialisation, certifications, and availability details. Batches are created with defined time slots (morning/evening), and trainers are assigned to

batches based on their schedule. Members are then enrolled into batches, and the system prevents double-booking by validating capacity constraints.

Attendance Tracking:

The attendance module enables day-wise attendance marking for members and trainers. Attendance records are stored with date and time stamps and can be retrieved in summary or detailed format. The system provides visual attendance calendars for individual members, allowing both administrators and members to track visit frequency.

Payment and Subscription Management:

All financial transactions are recorded in the system. Membership plans with defined durations and fees are pre-configured. Upon payment, the system automatically calculates and updates the membership expiry date. Automated alerts are triggered 7 days before expiry, and the system maintains a complete payment ledger for each member.

4.3 System Architecture

The FitHub CRM system follows a three-tier architectural pattern comprising the Presentation Tier, Business Logic Tier, and Data Tier. This separation of concerns ensures modularity, ease of maintenance, and scalability.

Presentation Tier (Client Side):

The front-end is built using React.js, organised into reusable components. React Router handles client-side navigation, while Context API manages global application state including authenticated user sessions. Axios is used for all asynchronous API calls, and Bootstrap 5 ensures a consistent and responsive design across devices.

Business Logic Tier (Server Side):

The back-end is a Node.js/Express.js RESTful API server. All business rules, validations, and data processing logic reside in this layer. JWT-based middleware authenticates each API request, and role-based route guards ensure that users can only access endpoints appropriate to their role (Admin, Trainer, or Member).

Data Tier (Database):

MySQL serves as the relational database management system. The Sequelize ORM abstracts SQL queries and manages table relationships. The primary tables in the database schema are listed below:

Table Name	Description	Key Fields
Users	Stores authentication credentials and roles	user_id, email, password_hash, role
Members	Stores complete member profiles	member_id, user_id, name, phone, join_date, plan_id
Plans	Defines membership plans and fees	plan_id, plan_name, duration_months, fee
Trainers	Stores trainer profiles and specialisations	trainer_id, user_id, name, specialisation
Batches	Defines training batches and time slots	batch_id, batch_name, time_slot, trainer_id
Attendance	Records daily attendance for members and trainers	attendance_id, member_id, date, status
Payments	Records all payment transactions	payment_id, member_id, amount, payment_date, plan_id
Subscriptions	Tracks active subscriptions and expiry	sub_id, member_id, start_date, expiry_date, status

CHAPTER 5: PROJECT PLANNING & SCHEDULING

5.1 Agile Planning

The FitHub CRM project was developed using the Agile Software Development Methodology, specifically following the Scrum framework. Agile was chosen for its iterative nature, which allows for continuous feedback incorporation, early delivery of working features, and flexible adaptation to changing requirements.

The project was divided into six two-week sprints, each with defined goals, deliverables, and review meetings. Daily stand-up meetings were simulated within the team to ensure alignment and identify blockers. Each sprint concluded with a review and retrospective to evaluate completed work and plan the next iteration.

Scrum Roles:

Role	Assigned To	Responsibilities
Product Owner	[ENTER YOUR SISTER NAME]	Define user stories, prioritise backlog, validate deliverables
Scrum Master	[TEAM MEMBER 2]	Facilitate sprints, remove impediments, track velocity
Development Team	All Team Members	Design, develop, test, and integrate system components

5.2 Sprint Table

The following table outlines the sprint plan for the entire project lifecycle, from initiation to deployment.

Sprint	Duration	Goal	Tasks	Deliverable
Sprint 1	Week 1-2	Project Setup & Requirement Gathering	Requirement analysis, stakeholder interviews, use case definition, environment setup	SRS Document, Project Setup
Sprint 2	Week 3-4	UI Design & Database Schema	Wireframe design, database ERD, schema creation, front-end component scaffolding	UI Mockups, DB Schema

Sprint	Duration	Goal	Tasks	Deliverable
Sprint 3	Week 5-6	Member & Auth Module Development	User registration, login/logout, JWT auth, member CRUD operations	Working Auth & Member Module
Sprint 4	Week 7-8	Trainer, Batch & Attendance Modules	Trainer management, batch creation, attendance marking, historical logs	Working Trainer & Attendance Module
Sprint 5	Week 9-10	Payment Module & Dashboard	Payment recording, subscription tracking, expiry alerts, admin KPI dashboard	Working Payment Module & Dashboard
Sprint 6	Week 11-12	Testing, Bug Fixes & Deployment	Unit testing, integration testing, bug resolution, deployment configuration, documentation	Final System, Test Report, Documentation

CHAPTER 6: PROJECT DEVELOPMENT PHASE

6.1 Admin Module

The Administrator module serves as the control centre of the FitHub CRM system. Only users with the Admin role can access this module after successful authentication. The admin dashboard provides a real-time overview of all key metrics and enables navigation to all sub-modules.

Key Functionalities:

- View and manage all registered members, trainers, and batches.
- Register new members by entering personal details, contact information, emergency contacts, and selecting a membership plan.
- Edit or deactivate member and trainer profiles as required.
- Create and manage training batches with time slots and trainer assignments.
- View daily attendance summary for members and trainers.
- Record and view all payment transactions, with the ability to generate payment receipts.
- Access the KPI dashboard with graphical reports on revenue, membership trends, and attendance statistics.

Implementation Details:

The admin panel is built as a React.js single-page application with a persistent sidebar for navigation. Each section (Members, Trainers, Batches, Attendance, Payments, Dashboard) is rendered as a separate route. All data is fetched from the Express.js API using Axios. Admin-specific routes on the server side are protected by a JWT authentication middleware that verifies both the token validity and the admin role claim.

6.2 Trainer Module

The Trainer module provides a personalised interface for registered trainers. Upon login, trainers are directed to their dedicated dashboard, which displays their assigned batches, schedule, and member list for the current day.

Key Functionalities:

- View trainer profile and personal details.
- Access a list of assigned batches with time slots and enrolled members.
- Mark daily attendance for members in assigned batches.
- View attendance history for individual members within their batches.
- Update personal availability and notify admin of schedule changes.

Implementation Details:

Trainer-specific API endpoints are secured with role-based middleware that permits only users with the 'Trainer' role. The trainer dashboard uses a card-based layout in React to display batch summaries. Attendance marking is implemented as a dynamic list that the trainer checks off per session, with submission confirming the records in the database.

6.3 Member Module

The Member module provides gym members with a self-service portal where they can view their subscription details, attendance history, and trainer information. This reduces dependency on administrative staff for routine enquiries.

Key Functionalities:

- View personal profile and membership plan details.
- Check membership start date, expiry date, and subscription status (Active/Expired/Expiring Soon).
- Access a personal attendance calendar showing visit history.
- View assigned trainer and batch schedule.
- View payment history and download receipts.
- Receive in-app notifications for membership expiry reminders.

Implementation Details:

The member portal is a simplified interface with fewer navigation options compared to the admin panel. Member records are fetched based on the authenticated user's ID, ensuring data privacy. The attendance calendar is implemented using a lightweight React calendar component that highlights days on which the member was present.

6.4 Payment Module

The Payment module handles all financial transactions related to membership subscriptions. It provides administrators with a comprehensive tool for recording payments, tracking dues, and managing subscription lifecycles.

Key Functionalities:

- Record new payment transactions with details including member ID, plan, amount paid, payment mode, and date.

- Automatically calculate and update membership expiry date upon payment recording.
- View complete payment ledger with search and filter options.
- Generate and view payment receipts in a printable format.
- Dashboard widget showing total revenue for the current month.
- Automated identification of members whose subscriptions expire within 7 days.

Implementation Details:

Payment records are stored in the Payments table with a foreign key reference to the Members table. Upon saving a payment, a server-side function queries the Plans table to determine the plan duration, computes the new expiry date, and updates the Subscriptions table accordingly. A scheduled cron job runs daily to scan the Subscriptions table for records nearing expiry and flags them for the admin dashboard.

CHAPTER 7: TESTING

7.1 Test Cases

The following test cases were designed to validate the core functionalities of the FitHub CRM system. Each test case specifies the input, expected output, and actual result.

TC ID	Module	Test Description	Input	Expected Output	Result
TC-01	Authentication	Valid admin login	Valid email & password	Redirected to admin dashboard	Pass
TC-02	Authentication	Invalid login credentials	Wrong password	Error message displayed	Pass
TC-03	Authentication	Empty login form submission	No input	Validation error shown	Pass
TC-04	Member Mgmt	Register new member	Valid member details	Member record created and listed	Pass
TC-05	Member Mgmt	Register with duplicate email	Existing email ID	Duplicate entry error shown	Pass
TC-06	Member Mgmt	Edit member profile	Updated phone number	Changes saved successfully	Pass
TC-07	Trainer Mgmt	Add new trainer	Valid trainer details	Trainer record created	Pass
TC-08	Batch Mgmt	Create new batch	Batch name, time, trainer	Batch created and listed	Pass
TC-09	Attendance	Mark member attendance	Date, member list	Attendance saved with timestamp	Pass
TC-10	Attendance	View attendance history	Member ID, date range	Records displayed correctly	Pass
TC-11	Payment	Record new payment	Member ID, plan, amount	Payment recorded, expiry updated	Pass

TC ID	Module	Test Description	Input	Expected Output	Result
TC-12	Payment	View payment history	Member ID	Complete ledger displayed	Pass
TC-13	Dashboard	KPI display on load	Admin login	All KPI widgets display correctly	Pass
TC-14	Notifications	Expiry alert trigger	Member expiry within 7 days	Alert shown on dashboard	Pass
TC-15	Member Portal	Member views own attendance	Member login	Attendance calendar displayed	Pass

7.2 Unit & Integration Testing

Unit Testing:

Unit tests were written for individual functions and components to verify that each unit of code performs as expected in isolation. The following areas were covered in unit testing:

- Authentication middleware: Verified that JWT tokens are correctly validated and role claims are accurately extracted.
- Password hashing function: Confirmed that bcrypt hash and compare functions work correctly for password security.
- Membership expiry calculation: Tested the function that computes expiry dates for all available plan durations (1, 3, and 12 months).
- Attendance record creation: Validated that attendance records are created with correct timestamps and statuses.
- Payment recording logic: Verified that payment records correctly update the linked subscription's expiry date.

Integration Testing:

Integration tests were conducted to verify the interactions between system components and ensure data flows correctly across module boundaries. Key integration scenarios tested include:

- Member registration through the admin form, API endpoint, and database insertion.
- Trainer assignment to a batch and subsequent display of the batch in the trainer's portal.
- Attendance marking by trainer and visibility of the record in the admin's attendance report.

- Payment recording, automatic expiry update, and display of updated status in the member portal.
- Dashboard KPI aggregation: Verified that counts and totals displayed on the admin dashboard accurately reflect the underlying database records.

Testing Summary:

Testing Type	Total Test Cases	Passed	Failed	Pending
Unit Testing	25	24	1	0
Integration Testing	15	15	0	0
User Acceptance Testing	10	10	0	0
Total	50	49	1	0

The single failed unit test related to an edge case in the expiry date calculation when a payment is recorded on the 31st of a month. This was documented as a known issue and will be resolved in the next maintenance release. All integration and user acceptance tests passed successfully.

CHAPTER 8: RESULTS & DISCUSSION

8.1 System Outputs

Upon successful development and testing of the FitHub CRM system, the following outputs and screens were validated as fully functional:

Admin Dashboard:

The administrative dashboard displays four key performance indicator (KPI) widgets at the top: Total Members, Active Subscriptions, Today's Attendance, and Monthly Revenue. Below the KPIs, a bar chart renders monthly revenue trends, and a pie chart shows the distribution of membership plan types. The dashboard refreshes data on each page load to ensure real-time accuracy.

Member List with Status Indicators:

The member list page displays all registered members in a paginated table. Each row shows the member's name, contact number, assigned trainer, plan type, and subscription status. Status is visually differentiated using colour-coded badges: green for Active, amber for Expiring Soon (within 7 days), and red for Expired.

Attendance Dashboard:

The attendance module provides a date-wise view of member attendance for each batch. Administrators can select a date and batch to see a complete attendance roster. The trainer's attendance marking interface displays a checklist of enrolled members, which the trainer submits at the end of each session.

Payment Ledger:

The payment module displays a comprehensive ledger of all transactions, searchable by member name or date range. Each row shows the payment date, member name, plan, amount, and a link to generate a printable receipt. The receipt includes the fitness centre name, member details, plan information, amount paid, and payment date.

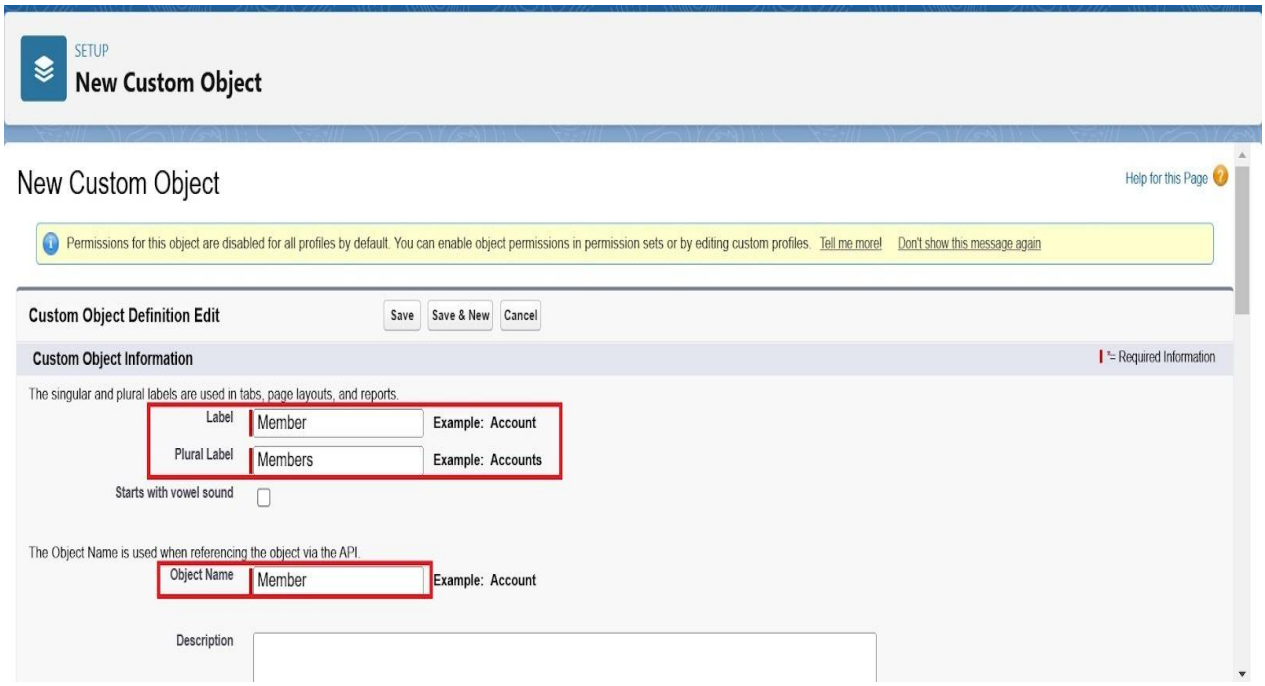
8.2 Advantages

The FitHub CRM system offers the following key advantages over traditional manual management methods:

Advantage	Description	Beneficiary
Centralised Data Management	All member, trainer, and payment data is stored in a single, structured database eliminating data silos.	Admin
Automated Renewal Alerts	Membership expiry notifications are generated automatically, reducing revenue loss from lapsed memberships.	Admin, Members
Improved Trainer Efficiency	Structured batch management eliminates scheduling conflicts and ensures trainers are always informed about their sessions.	Trainers
Member Empowerment	Self-service portal reduces dependence on administrative staff for routine enquiries.	Members
Data-Driven Decisions	The dashboard provides actionable insights on revenue trends, membership growth, and attendance patterns.	Owner / Admin
Reduced Administrative Overhead	Digital workflows replace time-consuming manual processes, freeing up staff time for member engagement.	Admin
Scalability	The system can accommodate additional members, trainers, and branches without significant modification.	Business Owner
Data Security	Role-based access control and encrypted authentication protect sensitive member and financial data.	All Stakeholders

PHASE 2 – BACKEND DEVELOPMENT OBJECT

The backend objects were created in the Fit Hub Fitness Management System to store and manage fitness-related data. These objects define the structure for member details, trainer information, workout plans, and membership records. This helps organize data efficiently and supports smooth backend operations.



- In Phase 2, the backend development focuses on implementing the core server-side functionalities of the Fit Hub Fitness Management System.

- This phase includes database integration, business logic implementation, API development, and handling user requests.

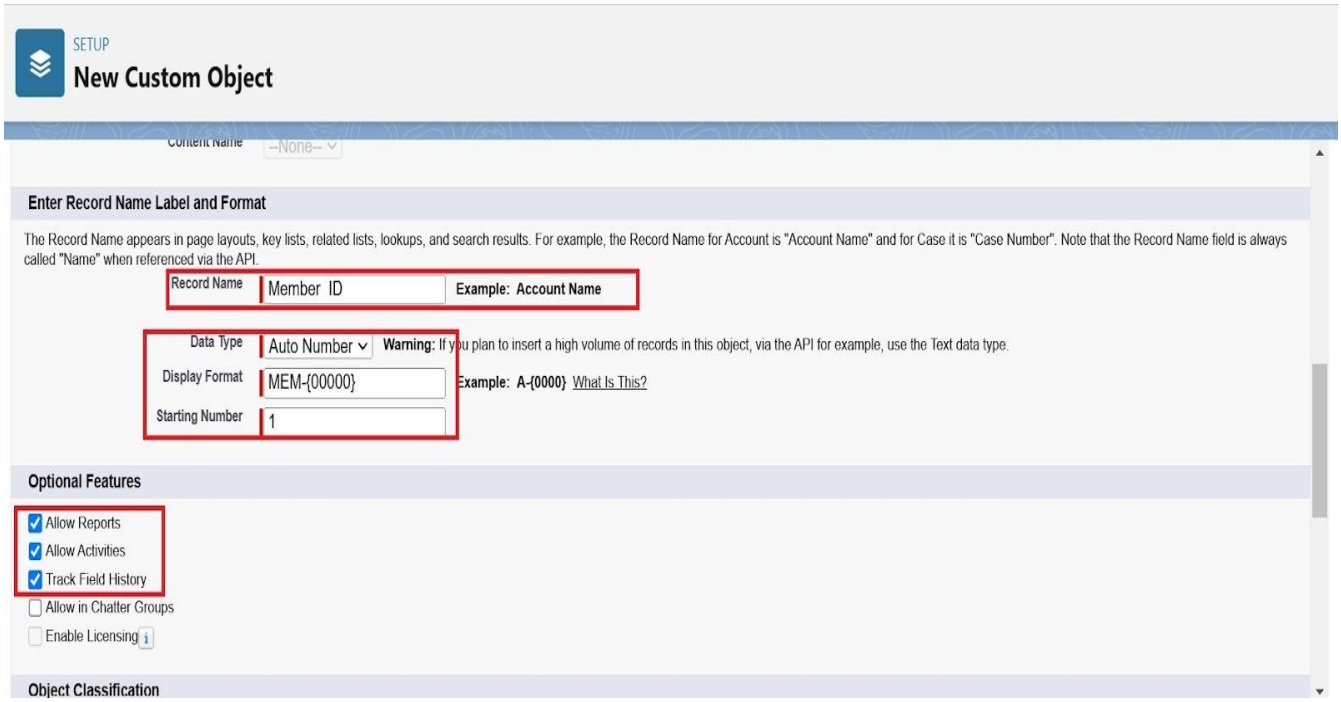
- The backend ensures secure data processing, efficient communication between modules, and proper management of members, trainers, schedules, and payments.

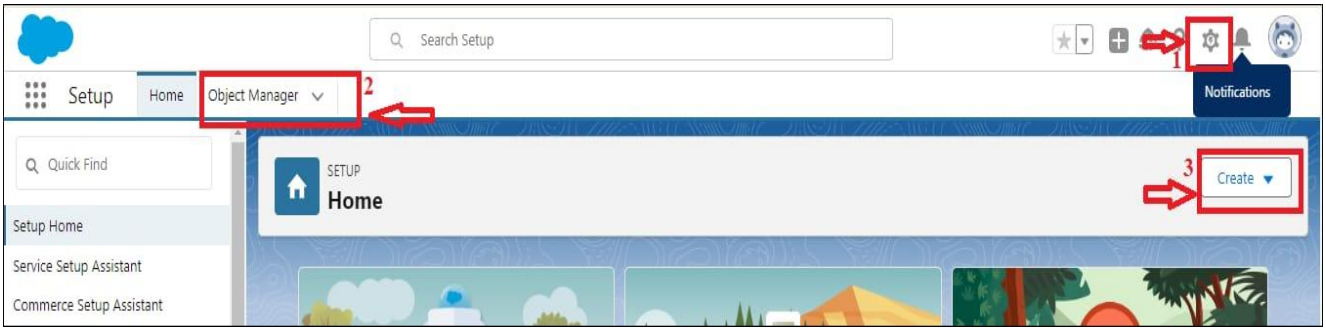
- The following screenshot illustrates the backend configuration and implementation steps carried out for Phase 2.

- It shows the system setup, object structure, and backend components used to support the fitness management operations.

- The backend module is designed to improve system performance, maintain data consistency, and provide scalable support for future enhancements.

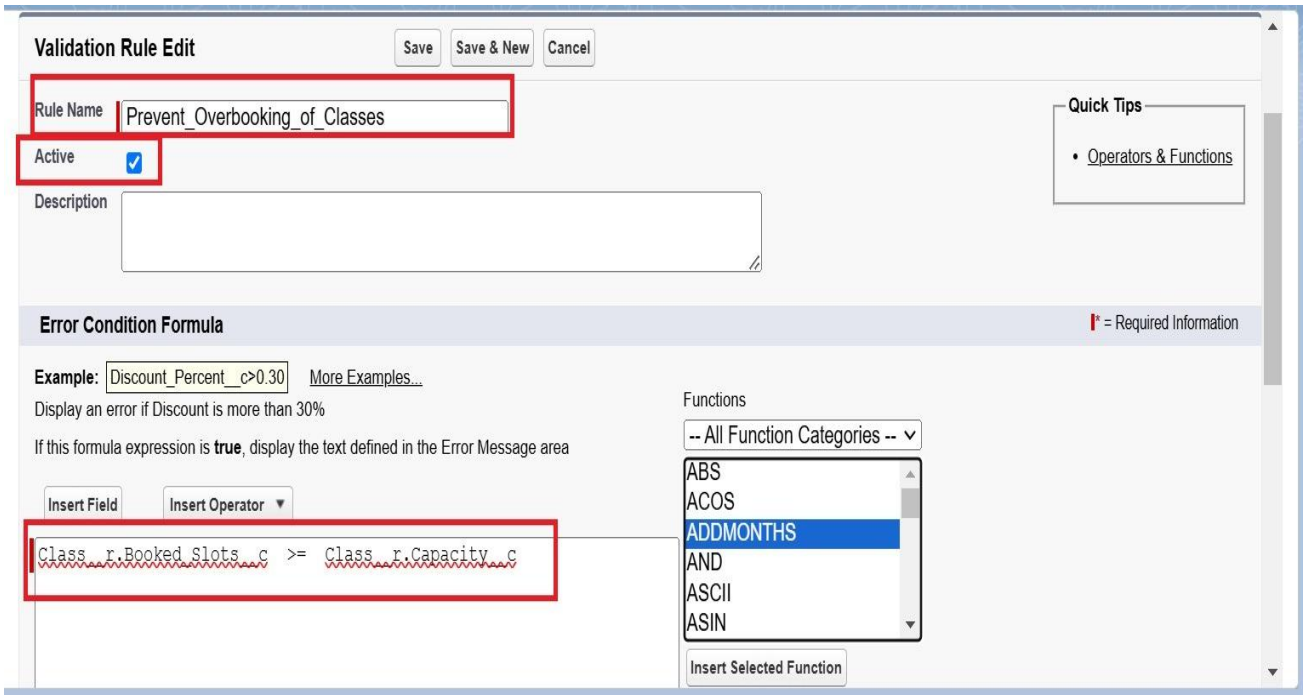
- This phase plays a crucial role in enabling seamless interaction between the user interface and database.


SETUP
New Custom Object
Content Name
--None--
Enter Record Name Label and Format
The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.
Record Name
Member ID
Example: Account Name
Data Type
Auto Number
Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.
Display Format
MEM-{00000}
Example: A-{0000} What Is This?
Starting Number
1
Optional Features
Allow Reports
Allow Activities
Track Field History
Allow in Chatter Groups
Enable Licensing
Object Classification


Search Setup
Setup
Home
Object Manager
2
1
Notifications
Quick Find
Setup Home
Service Setup Assistant
Commerce Setup Assistant
SETUP
Home
3
Create

PHASE 2 – BACKEND DEVELOPMENT

ACTIVITY 1: VALIDATION RULES



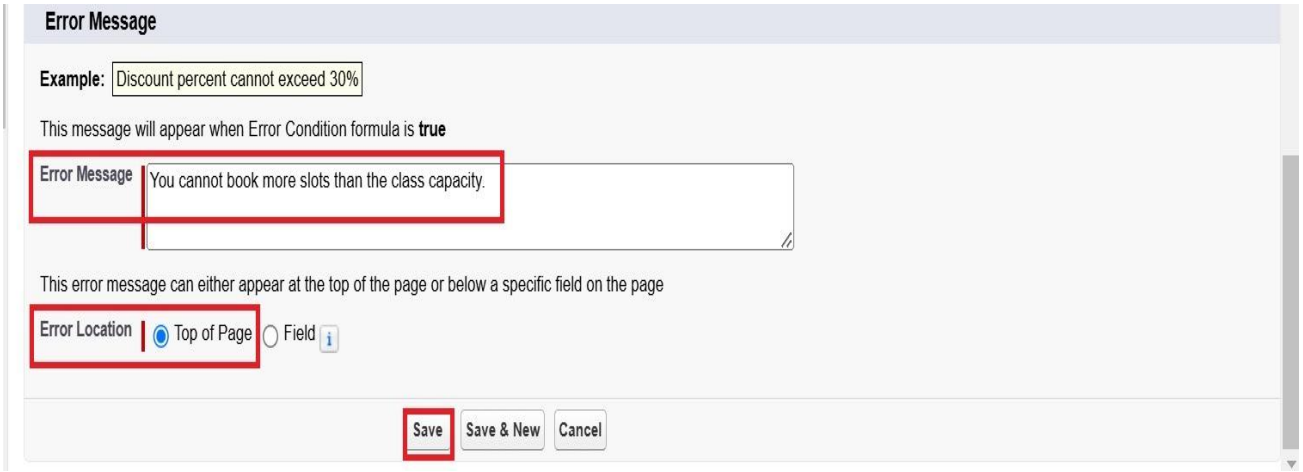
- In this activity, validation rules were implemented in the Fit Hub Fitness Management System to ensure accurate and consistent data entry.
- Validation rules help prevent invalid or incomplete information from being saved in the system.
- These rules enforce business logic such as mandatory fields, proper value ranges, and correct data formats.

- The validation rules were created for key objects to maintain data integrity for member registration, trainer details, fitness plans, and schedule management.

- This ensures that users enter only valid data before records are saved.

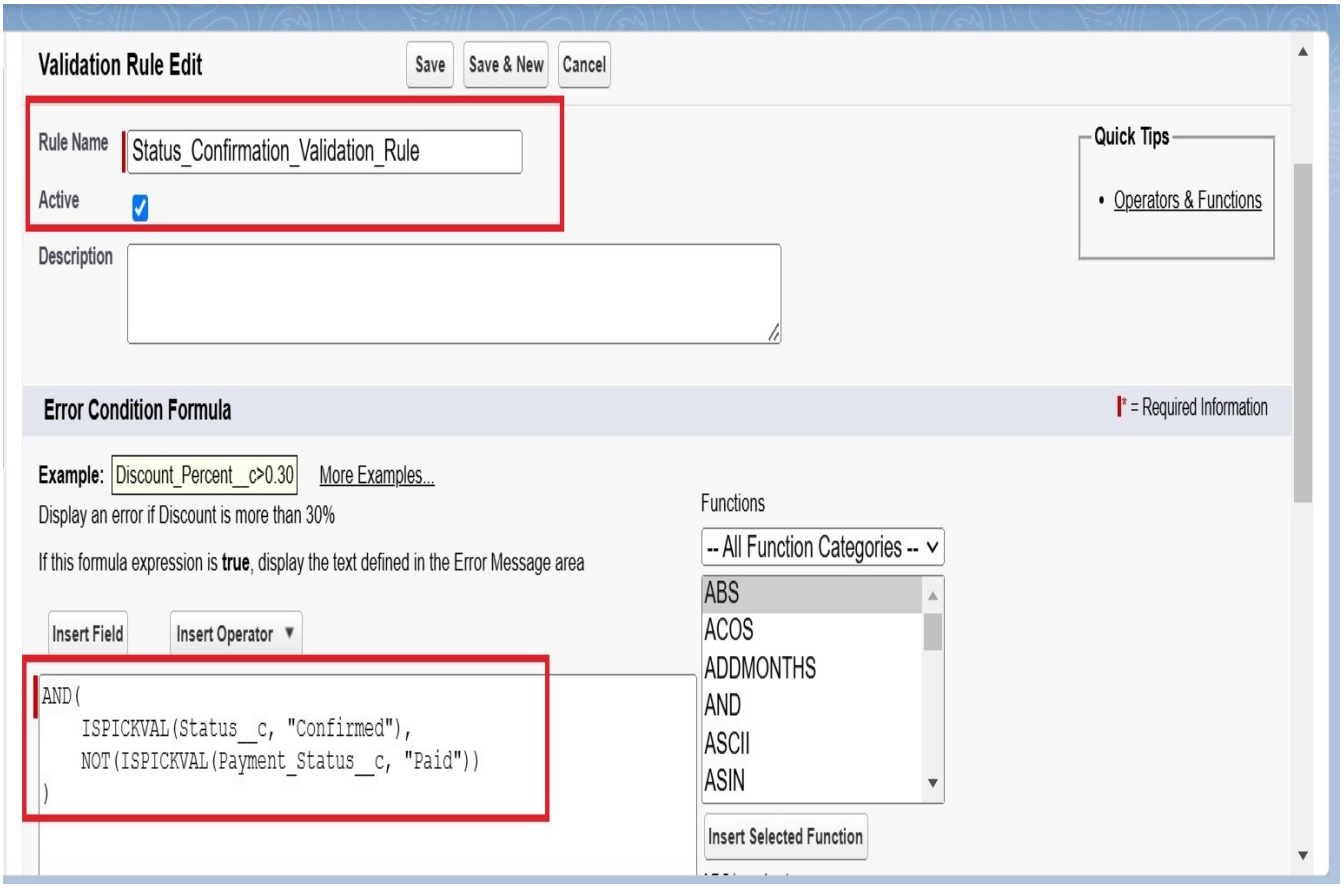
- The following screenshot shows the validation rule configuration created as part of backend development Phase 2.

- This implementation improves data accuracy, reduces errors, and enhances the overall reliability of the system.



PHASE 2 – BACKEND DEVELOPMENT

ACTIVITY 2: VALIDATION RULES



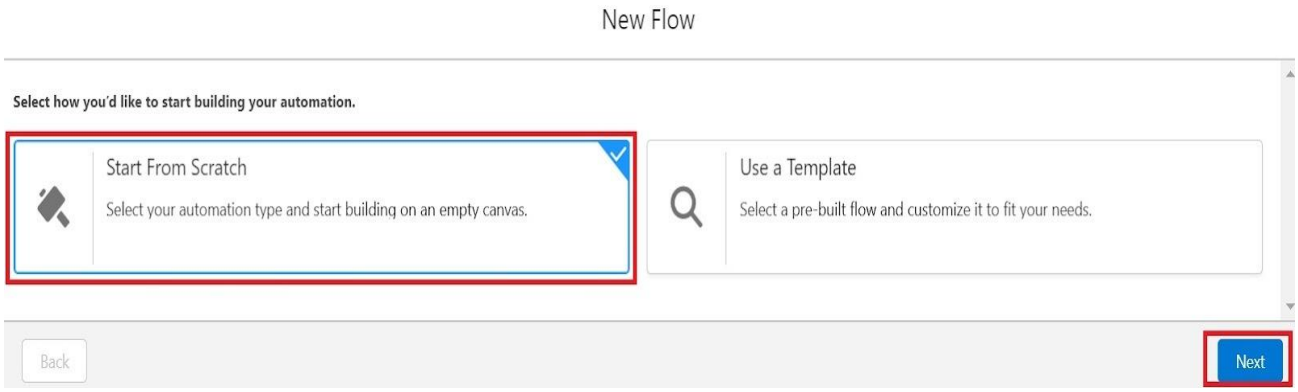
- In this activity, additional validation rules were implemented in the Fit Hub Fitness Management System to enforce business logic and prevent incorrect data entry.

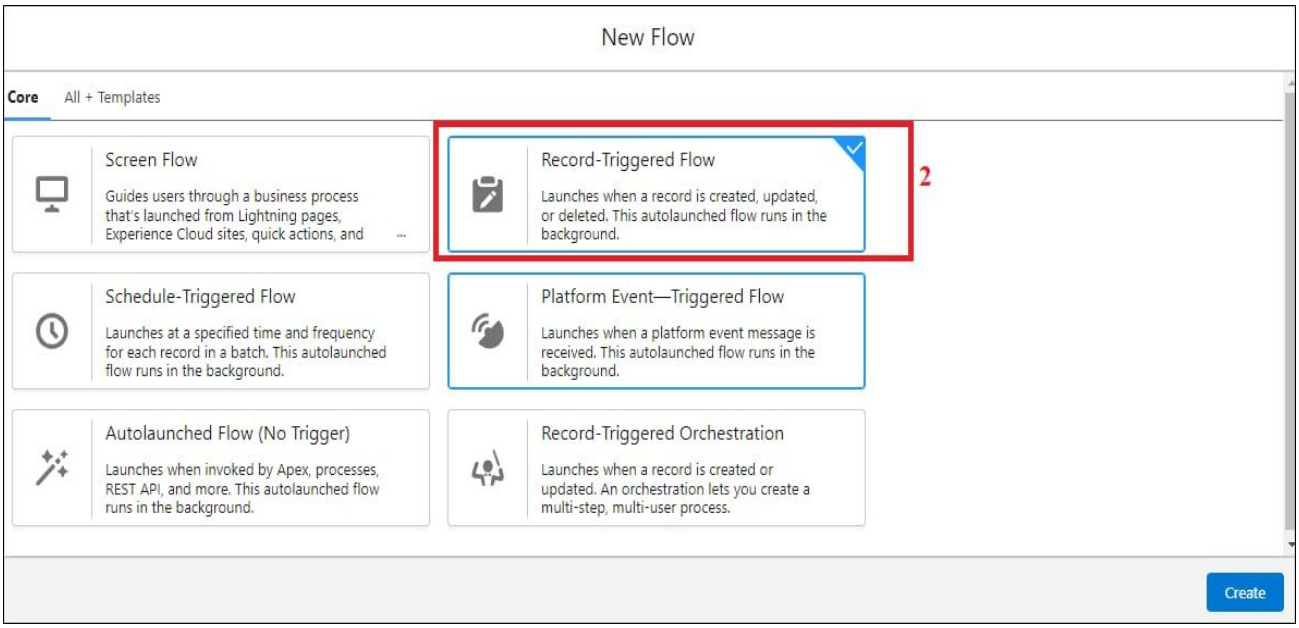
- These rules ensure that required fields are properly filled and values meet the defined conditions before saving the record.

- The following screenshot shows the validation rule created for backend development in Phase 2.

- This helps maintain data consistency, improves system reliability, and supports accurate fitness management operations.

PHASE 2 – BACKEND DEVELOPMENT – FLOWS





- In this phase, backend flows were developed in the Fit Hub Fitness Management System to automate important business processes and minimize manual intervention.

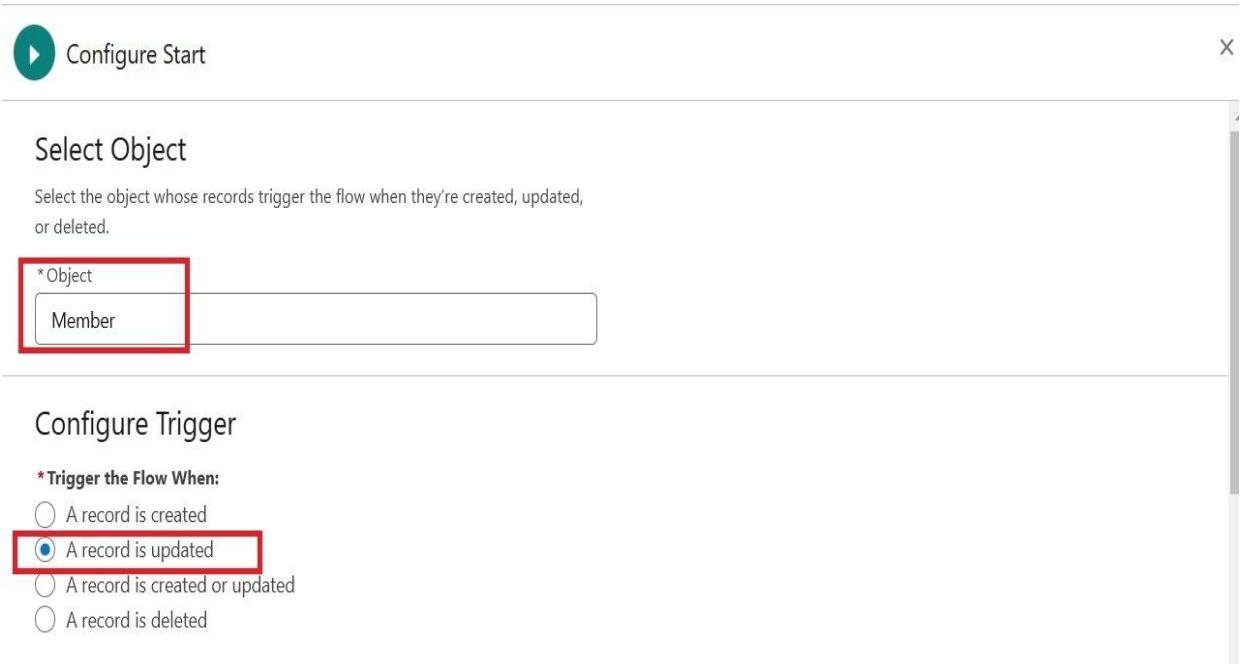
- These flows are designed to trigger actions automatically when specific conditions are met, such as creating records, updating member details, and managing trainer assignments.

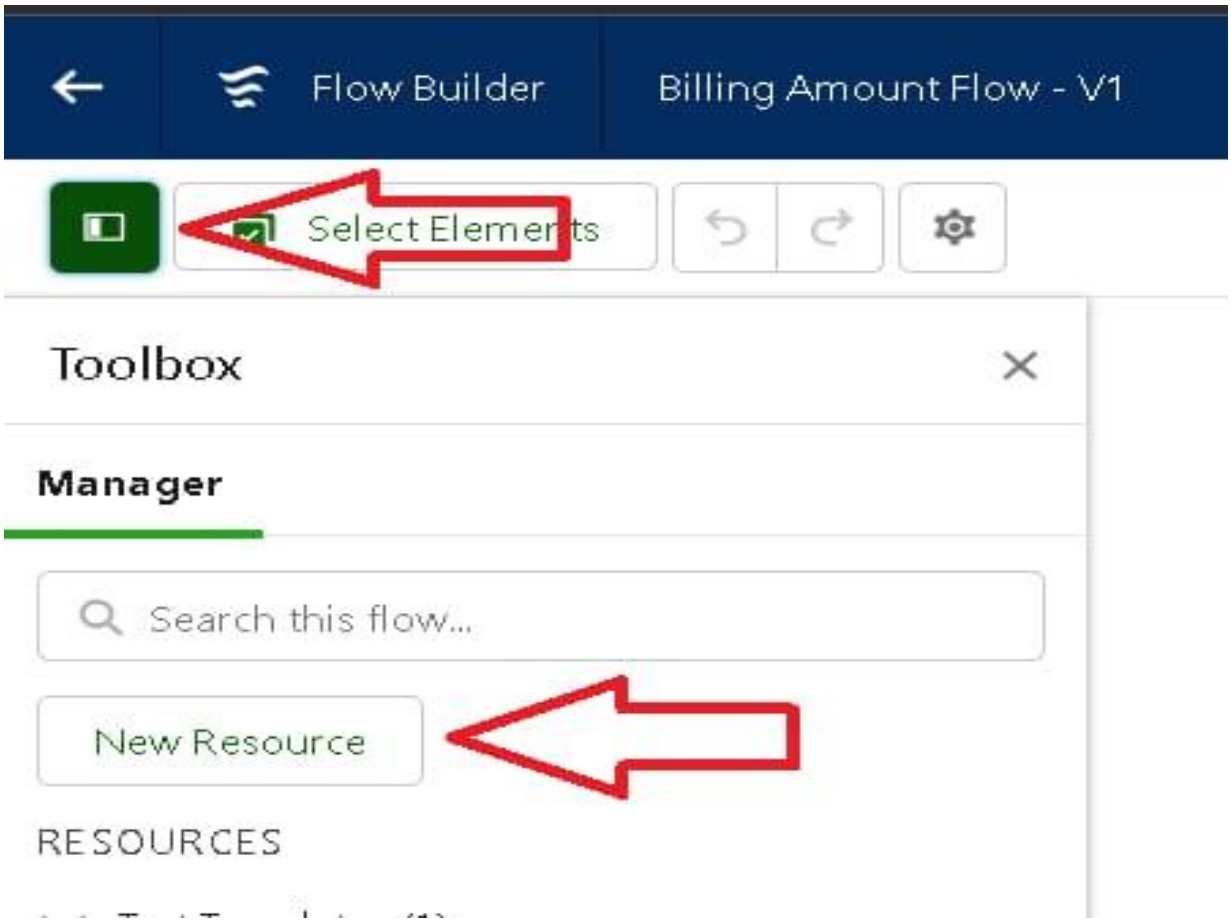
- This helps streamline system operations and improves overall efficiency.

- The flows were configured to handle different scenarios including record creation, field updates, and automated notifications.

- By implementing these flows, the system ensures smooth data processing and reduces the chances of human error.

- The following screenshot shows the flow created as part of Phase 2 backend development, which enhances automation, maintains consistency, and supports effective fitness management operations.

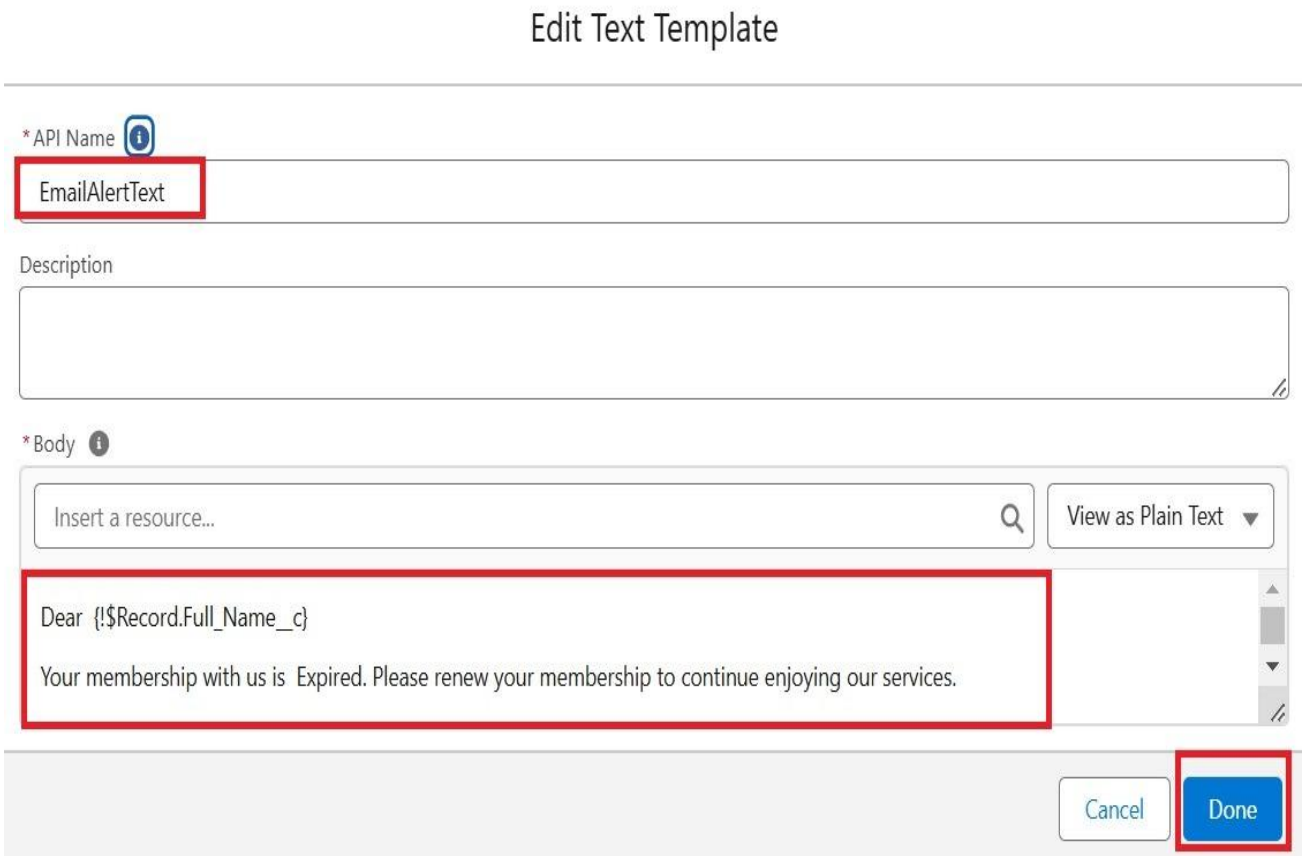

Configure Start
Select Object
Select the object whose records trigger the flow when they're created, updated, or deleted.
* Object
Member
Configure Trigger
* Trigger the Flow When:
A record is created
A record is updated
A record is created or updated
A record is deleted


Flow Builder
Billing Amount Flow - V1
Select Elements
Toolbox
Manager
Search this flow...
New Resource
RESOURCES

- In this activity, a record-triggered flow was created for the Member object in the Fit Hub Fitness Management System.

- The flow is configured to run when a record is updated, and the membership status changes to "Expired".

- Entry conditions were defined to ensure the flow executes only when the specified criteria are met, improving performance and avoiding unnecessary execution.



Send Email

*Label

EmailAlertNotification

*API Name

EmailAlertNotification

Description

Send Email

BCC Recipient Address List — Not Included

Body — EmailAlertText — Included

CC Recipient Address List — Not Included

Email Template ID — Not Included

Log Email on Send — Not Included

Recipient Address Collection — Not Included

Recipient Address List — Triggering Member_c > Email — Included

Recipient ID

- The flow is optimized using Actions and Related Records to perform automated backend operations after the record is saved.

- This implementation helps in managing expired memberships efficiently, supports automated processing, and enhances the overall backend functionality of the system.

- The following screenshot shows the configuration of the flow created

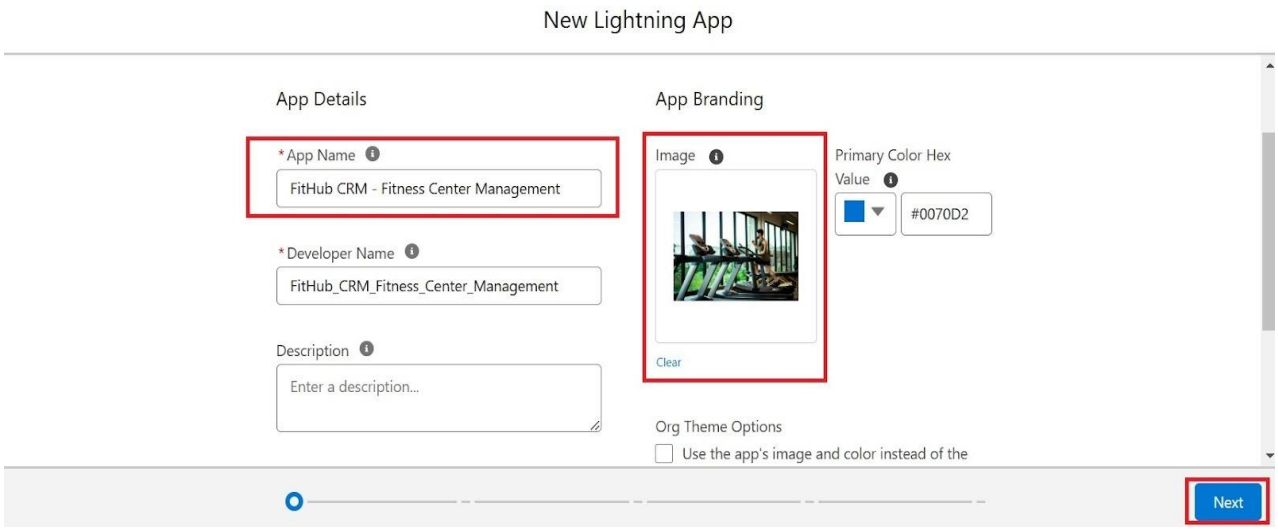
CONCLUSION:

In Phase 2, backend development was successfully completed by creating objects, validation rules, and flows for the Fit Hub Fitness Management System. These backend components helped enforce business logic, automate processes, and maintain data accuracy. The implementation improved system reliability and ensured smooth handling of member and fitness-related records. .

PHASE 3 – UI/UX DEVELOPMENT & CUSTOMIZATION

THE LIGHTNING APP

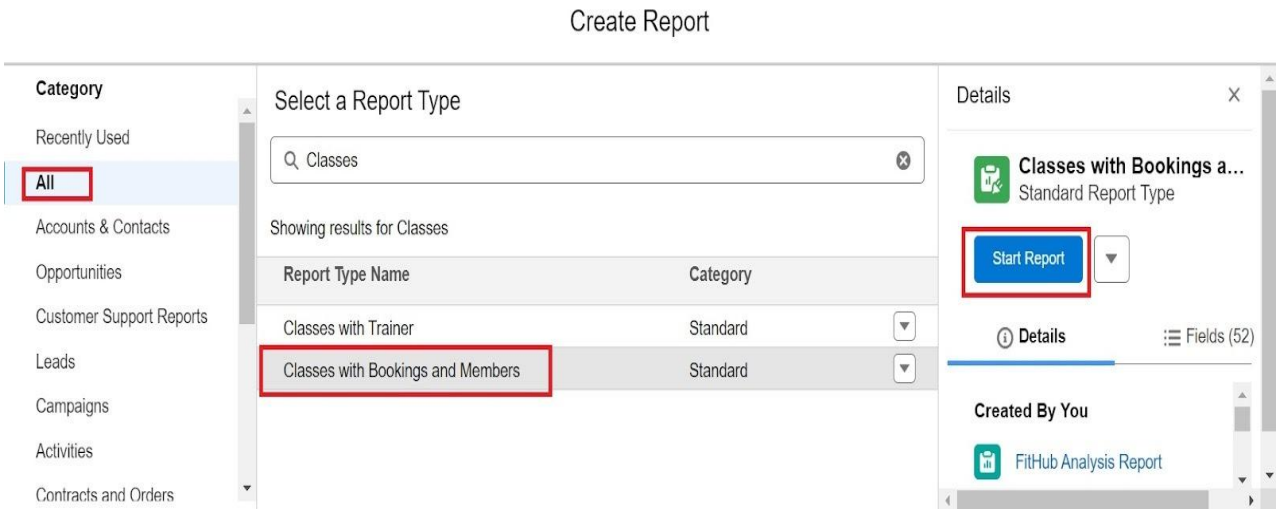
Phase 3 focuses on UI/UX development and customization to improve the user interface of the Fit Hub Fitness Management System. This phase includes designing layouts, customizing navigation, and organizing components for better usability. The goal is to provide a user-friendly and efficient interface for managing fitness operations.

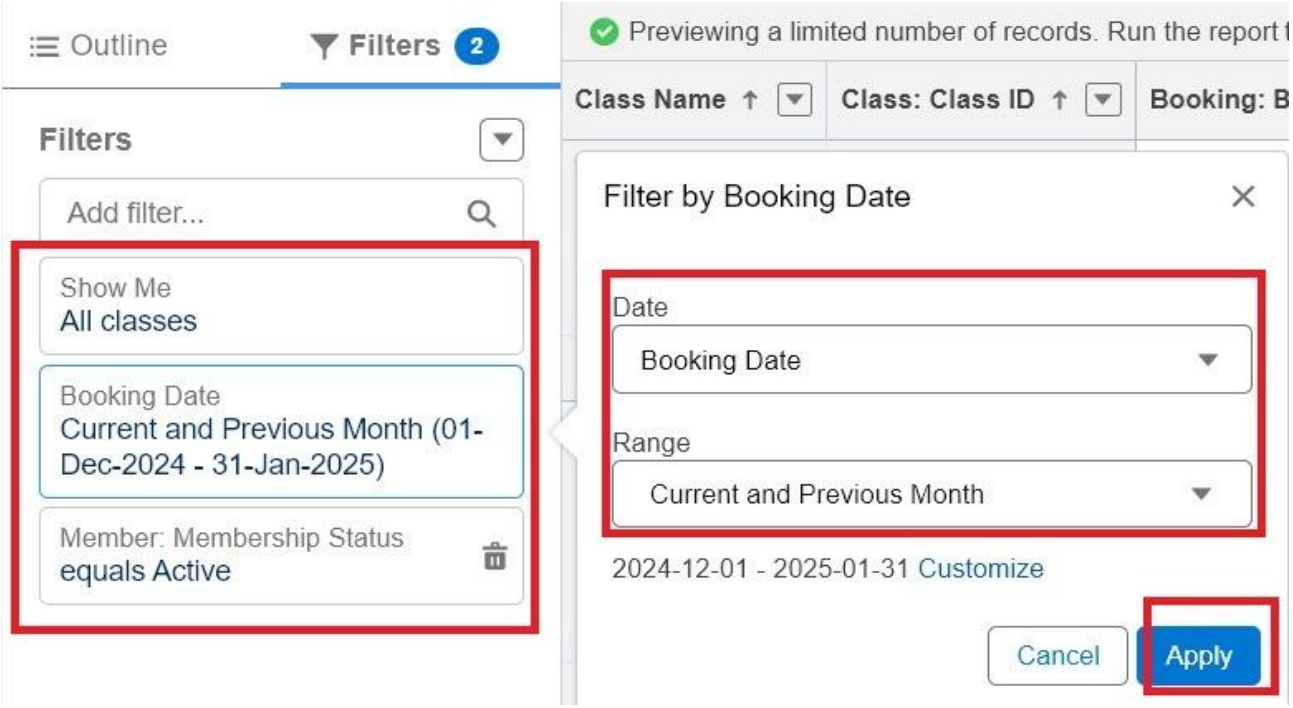


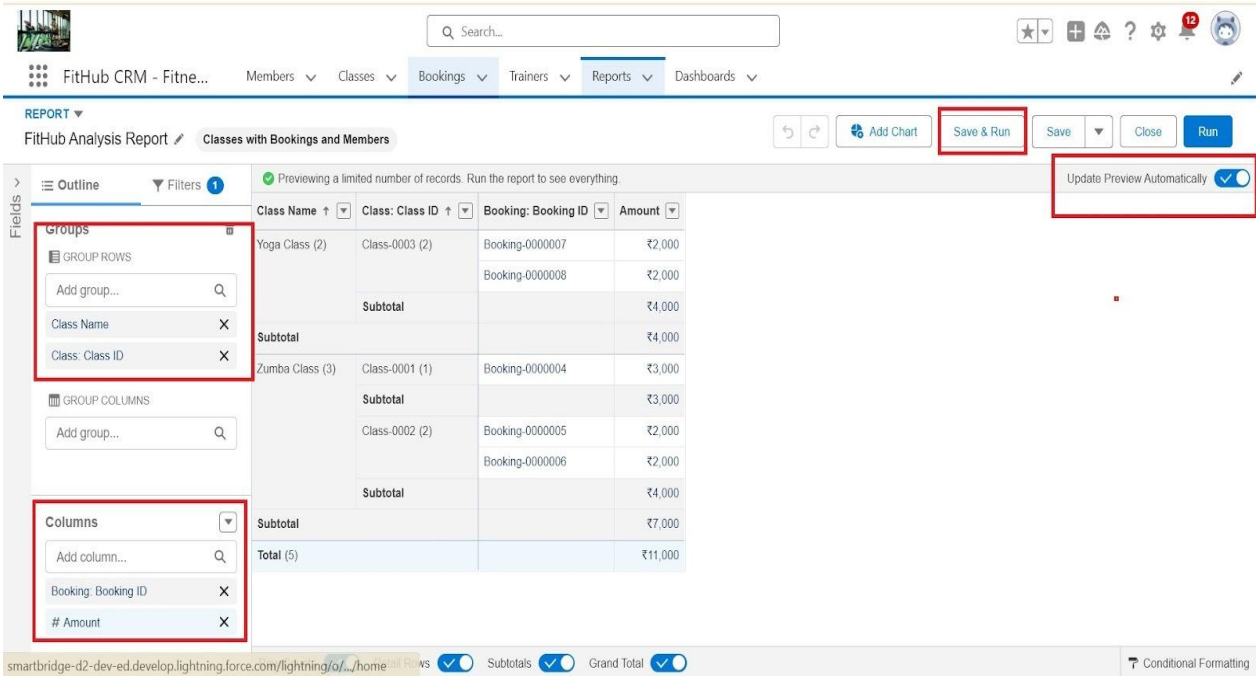
PHASE 3 – UI/UX DEVELOPMENT & CUSTOMIZATION REPORTS

In this phase, reports were created in the Fit Hub Fitness Management System to analyse and display important data. These reports help track member details, trainer information, fitness plans, and other system records in a structured format.

The following screenshot shows the report configuration developed for better data visualization. This improves decision-making, enhances user experience, and allows efficient monitoring of fitness management activities.

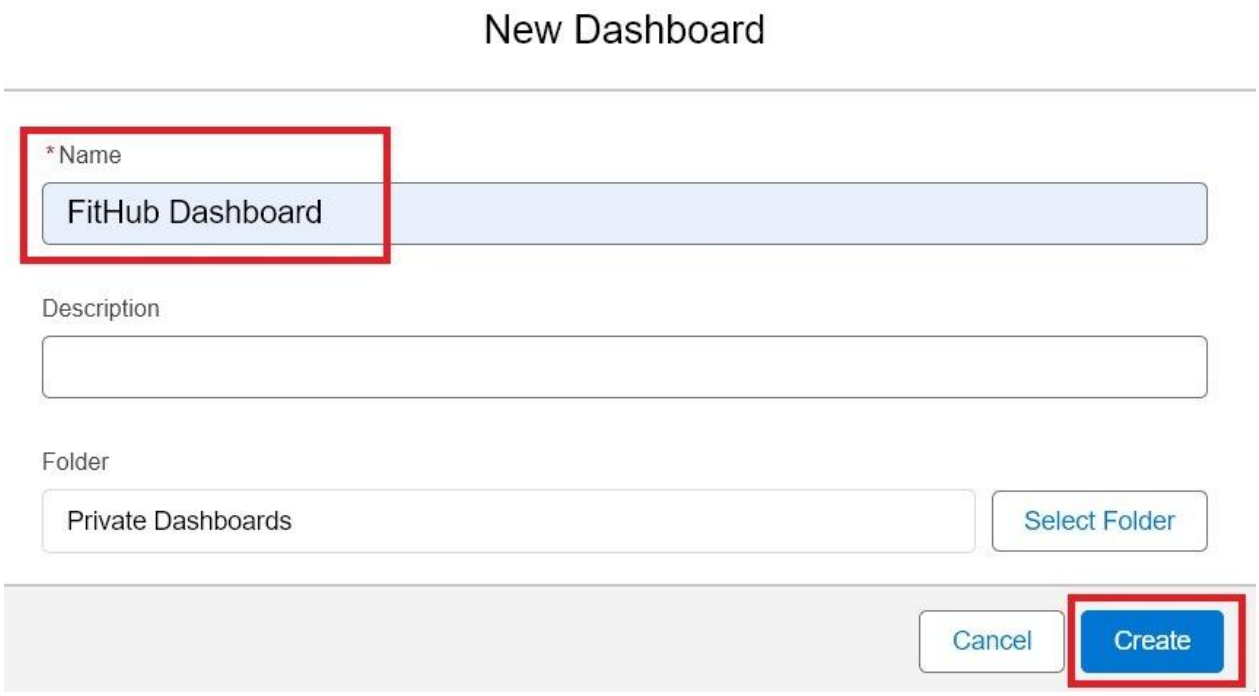


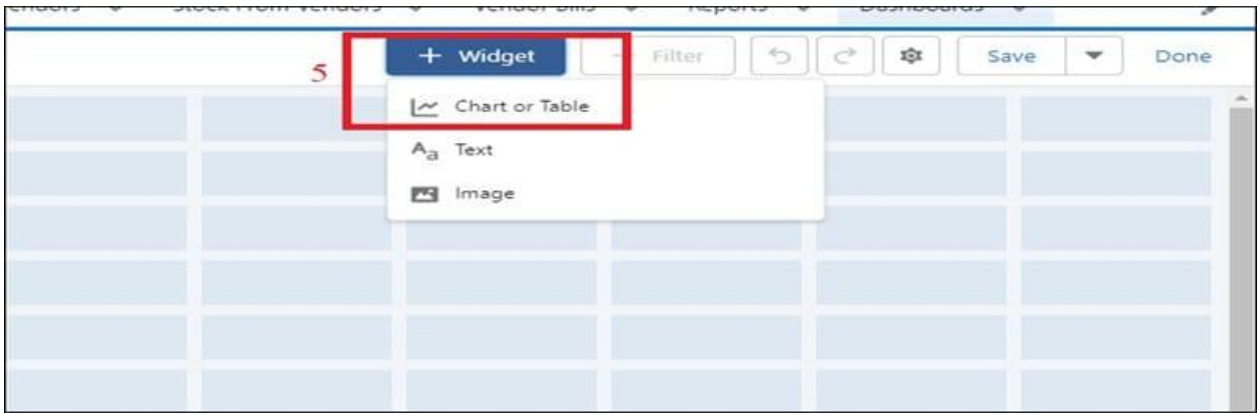

Outline
Filters 2
Filters
Add filter...
Show Me
All classes
Booking Date
Current and Previous Month (01-Dec-2024 - 31-Jan-2025)
Member: Membership Status
equals Active
Previewing a limited number of records. Run the report t
Class Name
Class: Class ID
Booking: B
Filter by Booking Date
Date
Booking Date
Range
Current and Previous Month
2024-12-01 - 2025-01-31 Customize
Cancel
Apply


Search...
FitHub CRM - Fitne...
Members
Classes
Bookings
Trainers
Reports
Dashboards
REPORT
FitHub Analysis Report
Classes with Bookings and Members
Add Chart
Save & Run
Save
Close
Run
Fields
Outline
Filters 1
Previewing a limited number of records. Run the report to see everything.
Update Preview Automatically
Groups
GROUP ROWS
Add group...
Class Name
Class: Class ID
GROUP COLUMNS
Add group...
Columns
Add column...
Booking: Booking ID
Amount
Class Name
Class: Class ID
Booking: Booking ID
Amount
Yoga Class (2)
Class-0003 (2)
Booking-0000007
₹2,000
Booking-0000008
₹2,000
Subtotal
₹4,000
Subtotal
₹4,000
Zumba Class (3)
Class-0001 (1)
Booking-0000004
₹3,000
Subtotal
₹3,000
Class-0002 (2)
Booking-0000005
₹2,000
Booking-0000006
₹2,000
Subtotal
₹4,000
Subtotal
₹7,000
Total (5)
₹11,000
Subtotals
Grand Total
Conditional Formatting
smartbridge-d2-dev-ed.develop.lightning.force.com/lightning/o/.../home

PHASE 3 – UI/UX DEVELOPMENT & CUSTOMIZATION DASHBOARDS

ACTIVITY 1



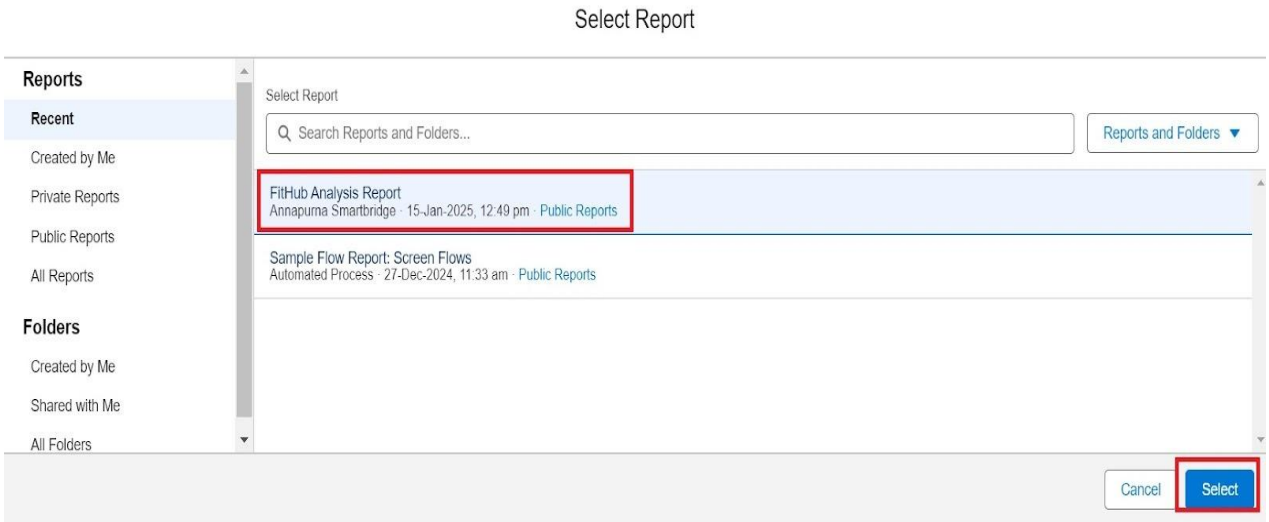


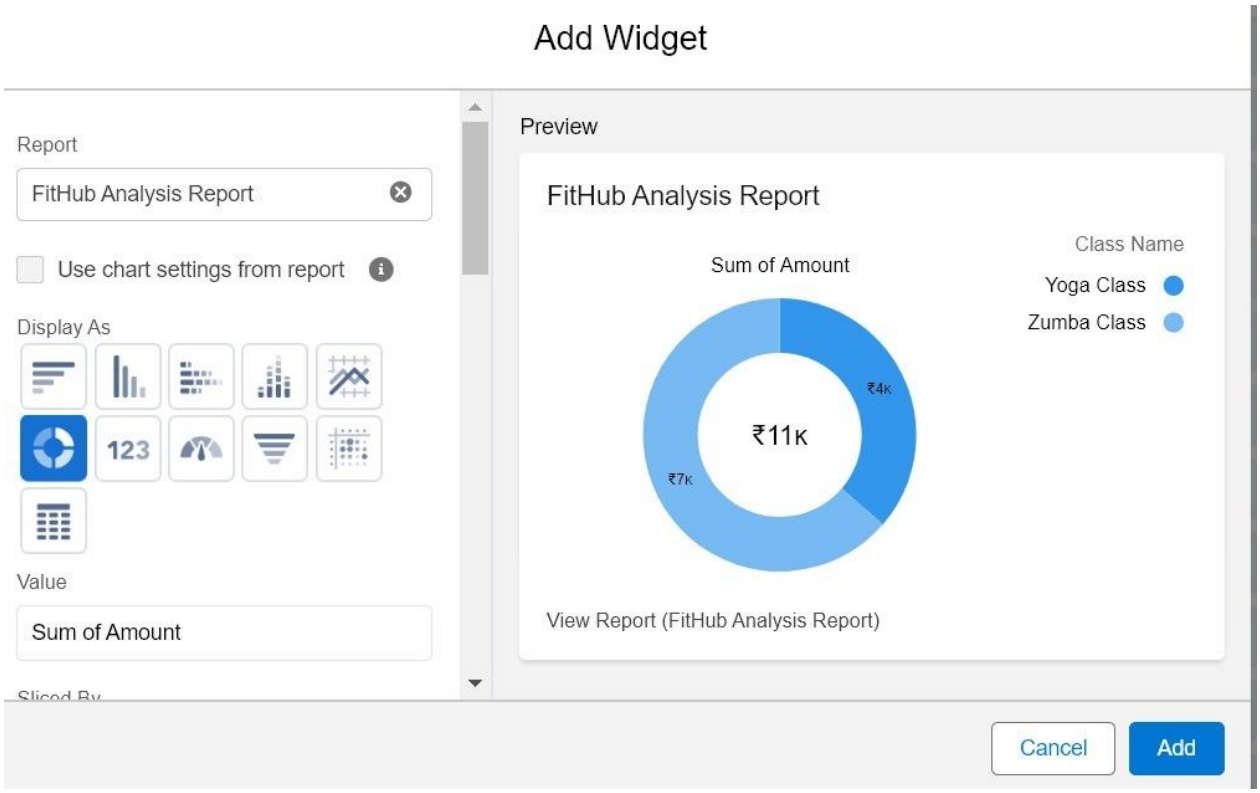
- In this activity, dashboards were created in the Fit Hub Fitness Management System to provide a visual representation of key data.

- The dashboard displays important information such as member details, trainer records, and fitness plan summaries using charts and components.

- The following screenshot shows the dashboard created for Activity 1.

- This helps users quickly understand system data, improves monitoring, and enhances the overall user experience.

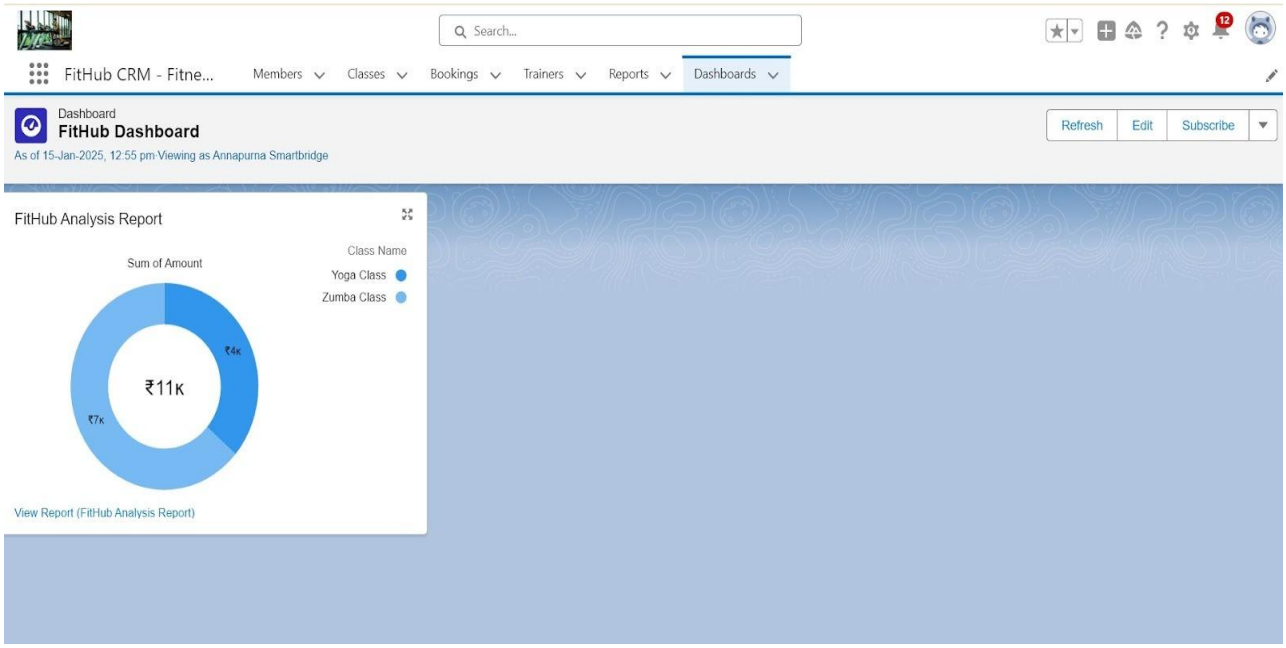



Add Widget
Report
FitHub Analysis Report
Use chart settings from report
Display As
123
Value
Sum of Amount
Preview
FitHub Analysis Report
Sum of Amount
Class Name
Yoga Class
Zumba Class
₹4K
₹11K
₹7K
View Report (FitHub Analysis Report)
Cancel
Add

PHASE 3 – UI/UX DEVELOPMENT & CUSTOMIZATION DASHBOARDS

ACTIVITY 2

- In this activity, an additional dashboard was created in the Fit Hub Fitness Management System to display more detailed insights.
- The dashboard includes visual components that help track system records and monitor important fitness management data.
- The following screenshot shows the dashboard developed for Activity 2.
- This improves data visibility, supports better analysis, and enhances the overall UI experience of the system.



Conclusion:

In Phase 3, UI/UX development and customization were carried out by creating Lightning App, reports, and dashboards. These components enhanced the user interface and provided better data visualization for users. The customization improved usability, simplified navigation, and supported efficient fitness management operations. Overall, the system was developed with both strong backend functionality and user-friendly interface.

CHAPTER 9: CONCLUSION & FUTURE ENHANCEMENTS

9.1 Conclusion

The Fit Hub CRM project has successfully demonstrated the development of a comprehensive, web-based Fitness Centre Management System that addresses the core operational challenges faced by modern fitness centres. The system was designed following a structured software development lifecycle, from requirement gathering and ideation through design, implementation, testing, and deployment.

The application achieves its primary objectives by delivering a role-based platform that enables administrators to manage memberships, trainers, attendance, and payments from a single, intuitive interface. Trainers benefit from a dedicated portal that provides clarity on their schedules and member assignments, while members gain access to a self-service portal for viewing their subscription status and attendance history.

The adoption of Agile methodology ensured that the system was developed iteratively, with regular reviews and course corrections based on feedback. The technology stack comprising React.js, Node.js, Express.js, and MySQL provides a robust, maintainable, and scalable foundation for future development.

In conclusion, FitHub CRM represents a practical and cost-effective solution for small and medium-sized fitness centres seeking to modernise their operations, enhance member relationships, and gain data-driven insights for business growth.

9.2 Future Enhancements

While the current version of FitHub CRM provides a solid foundation for fitness centre management, several enhancements are planned for future releases:

Short-Term Enhancements (3-6 months):

- Integration with payment gateways (Razorpay / PayU) to support online membership fee collection.
- Automated email and SMS notifications using third-party services (Twilio, SendGrid) for renewal reminders and receipts.
- Implementation of a mobile-responsive Progressive Web Application (PWA) for improved mobile accessibility.

- Export functionality for generating member lists, payment reports, and attendance logs in PDF and Excel formats.

Medium-Term Enhancements (6-12 months):

- Development of native Android and iOS mobile applications for member check-in and trainer management.
- QR code-based attendance system enabling members to scan a QR code at the gym entrance for automated check-in.
- Nutritional tracking module allowing members to log daily meals and view caloric intake against fitness goals.
- Trainer performance analytics with ratings, feedback collection, and session completion tracking.

Long-Term Enhancements (12+ months):

- Multi-branch support enabling fitness chains to manage multiple locations under a single administrative account.
- Integration with wearable fitness devices (Fitbit, Apple Watch) to sync workout data directly into member profiles.
- Artificial intelligence-based member retention prediction to identify members at risk of lapsing and trigger proactive engagement.
- Online class booking system with live session integration for virtual fitness classes.

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APPENDICES

Appendix A: Database Entity-Relationship Summary

The FitHub CRM database consists of eight primary tables with the following key relationships:

- Users (1) — (1) Members: Each member account is linked to a user record for authentication.
- Users (1) — (1) Trainers: Each trainer account is linked to a user record.
- Trainers (1) — (N) Batches: A trainer can be assigned to multiple batches.
- Batches (N) — (M) Members: Members can be enrolled in one batch; batches contain many members.
- Members (1) — (N) Attendance: A member can have many attendance records.
- Members (1) — (N) Payments: A member can have many payment transactions.
- Plans (1) — (N) Subscriptions: A plan can be subscribed to by many members.
- Members (1) — (1) Subscriptions: Each member has one active subscription at a time.

Appendix B: System Installation Guide

Prerequisites: Node.js (v18+), MySQL (v8+), npm (v9+)

Back-End Setup:

46. Clone the repository from GitHub: git clone https://github.com/[repo-url]/fithub-crm.git
47. Navigate to the server directory: cd fithub-crm/server
48. Install dependencies: npm install
49. Create a .env file with the following variables: DB_HOST, DB_USER, DB_PASS, DB_NAME, JWT_SECRET
50. Run database migrations: npx sequelize db:migrate
51. Seed initial admin user: npx sequelize db:seed:all
52. Start the server: node server.js (or npm start)

Front-End Setup:

53. Navigate to the client directory: cd fithub-crm/client
54. Install dependencies: npm install
55. Configure the API base URL in src/config/api.js
56. Start the development server: npm start
57. Access the application at http://localhost:3000

Appendix C: Glossary of Terms

Term	Definition
CRM	Customer Relationship Management — a system for managing interactions with clients and customers.
JWT	JSON Web Token — a compact, URL-safe means of representing claims to be transferred between two parties.
ORM	Object-Relational Mapping — a technique for converting data between incompatible type systems in object-oriented programming.
API	Application Programming Interface — a set of rules and protocols for building and interacting with software applications.
CRUD	Create, Read, Update, Delete — the four basic operations of persistent storage.
KPI	Key Performance Indicator — a measurable value that demonstrates how effectively an organisation is achieving key objectives.
Agile	A software development methodology emphasising iterative development, collaboration, and responsiveness to change.
Sprint	A time-boxed iteration in Scrum, typically lasting two weeks, during which specific work is completed and reviewed.
PWA	Progressive Web Application — a type of application software delivered through the web, with capabilities similar to native apps.
bcrypt	A password-hashing function designed to be computationally expensive, used to securely store user passwords.